July 12, 2023

Meet at **noon** tomorrow!

12pm-1pm : **Lecture**

1pm-2pm : **Studio**

Reflection 3 : Due Friday by midnight

Conceptual HW 3 : Due Sunday by midnight

Office Hours

Mondays Noon-2pm

Thursdays 2-3pm

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 2	July 3 No Class	July 4 No Classes!	July 5 V2 CH 5 HW	July 6 V2 CH 6 Studio : Fie alcs	July 7 V2 CH 6 Reflection
Week 3	July 10 V2 CH 6	July 11 Homework 2 Lab W3 - Usin	July 12 V2 CH 7	July 13 V2 CH 7 Studio : Complex E's	July 14 V2 CH 7 Reflection
Week 4	July 17 V2 CH 8	July 18 Homework 3 Lab W4 - Electric Potential	July 19 V2 CH 8	July 20 V2 CH 9 Studio : V and C	July 21 Reflection Quiz 1&2: CH 5-7

Chapter 6

Gauss's Law

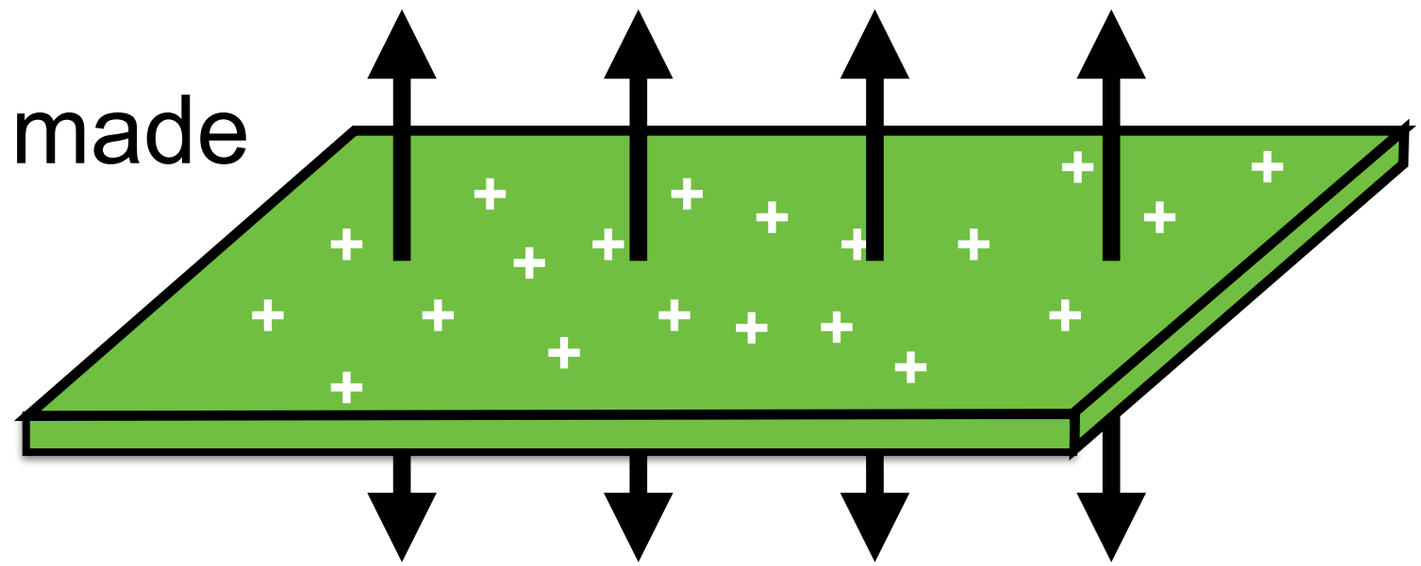
Using Gauss's Law, Again

§ 6.3
Lecture 08

Charged Plane

Field Prediction

What is the electric field made by an *infinite* plane with charge density η?



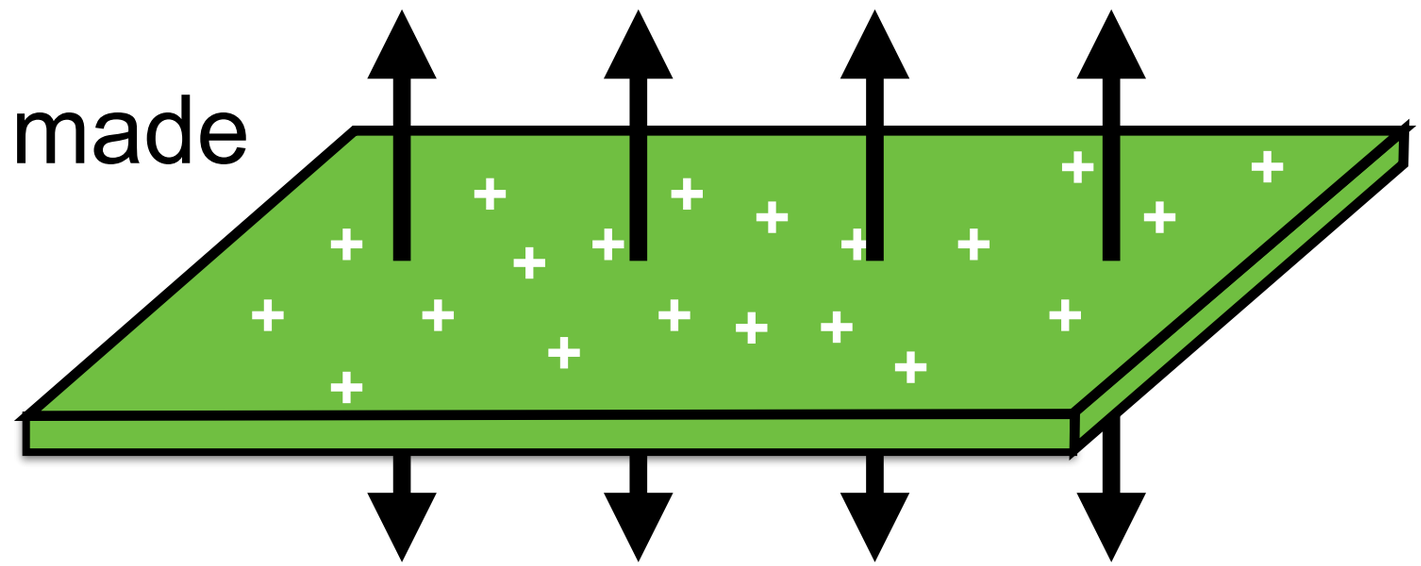
In your notes : what symmetries exist for the plane and what does it suggest the field might look like?

Rotate (in-plane)
Translate (along plane)
Mirror (intersecting plane)

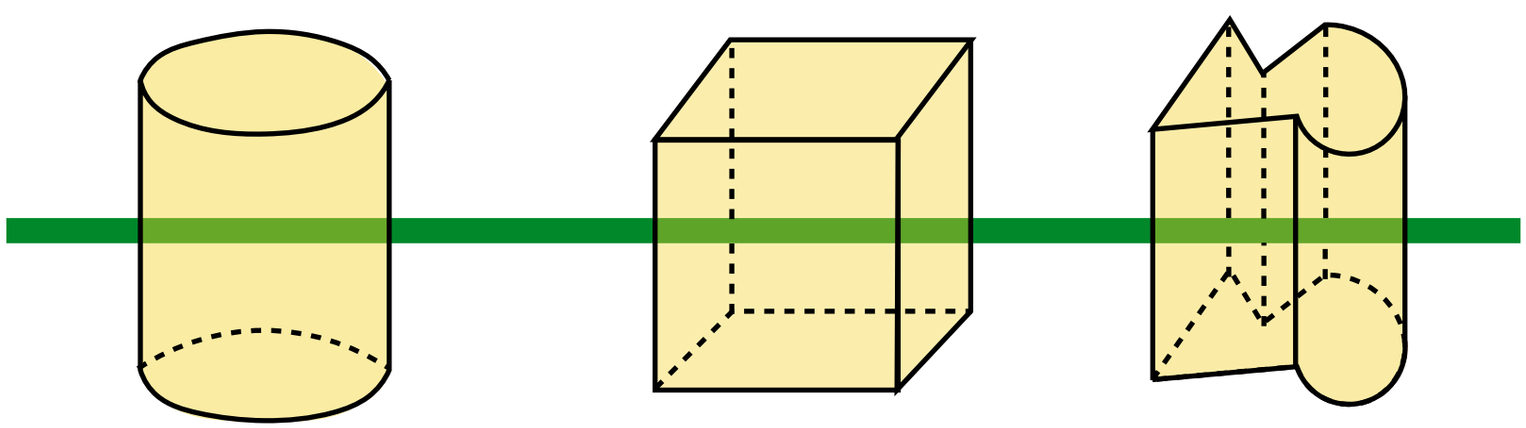
Charged Plane

Gaussian Surface

What is the electric field made by an *infinite* plane with charge density η?



In your notes : what type of surface could we use to make $\vec{E} \cdot d\vec{A}$ constant or zero along different sides?



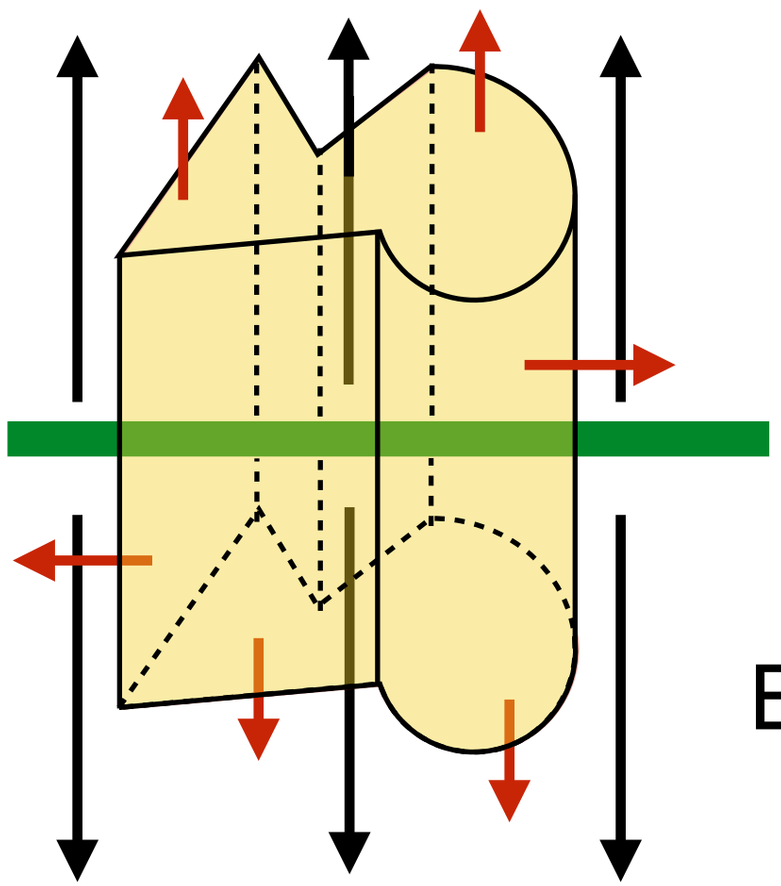
Radius : r
Height : H

Sides : L

Area : A
Height : H

Charged Plane

Flux Calculations



Area : A
Height : H

$$\text{Top}: \int \vec{E} \cdot d\vec{A} = \int E\, dA = E \int dA$$

$$= E\, A$$

$$\text{Bottom}: \int \vec{E} \cdot d\vec{A} = \int E\, dA = E \int dA$$

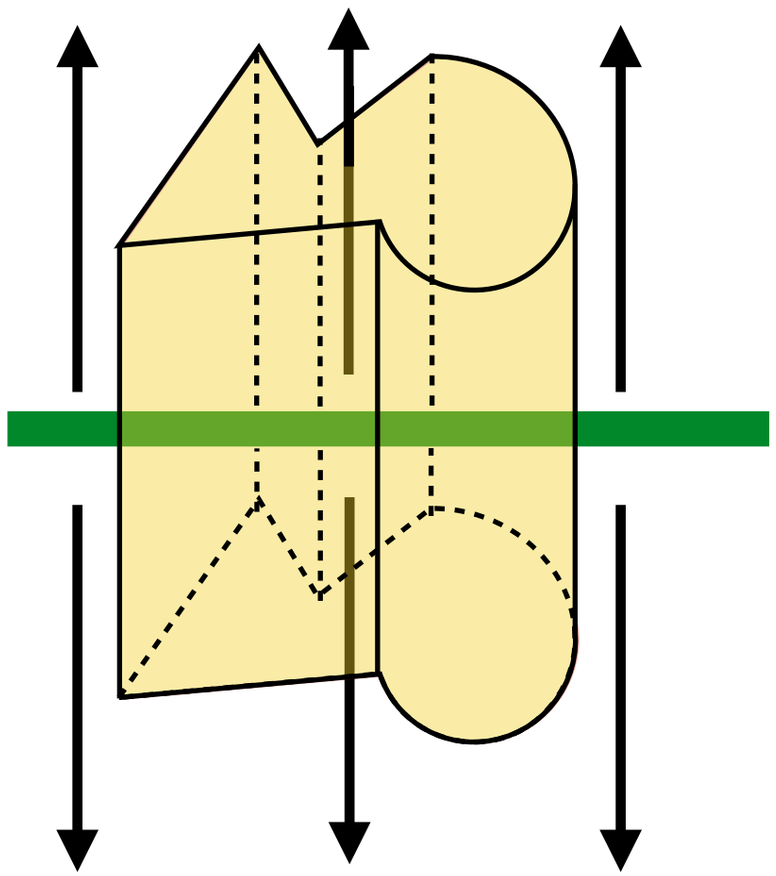
$$= E\, A$$

$$\text{Sides}: \int \vec{E} \cdot d\vec{A} = 0$$

$$\text{Total Flux}: \oint \vec{E} \cdot d\vec{A} = \Phi_{E,top} + \Phi_{E,bot} + \Phi_{E,sides} = 2\, E\, A$$

Charged Plane

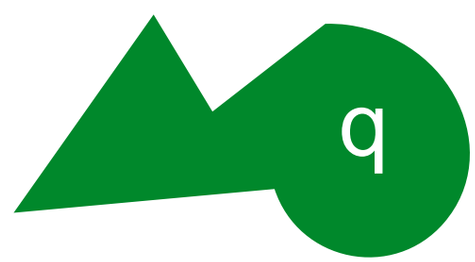
Gauss's Law



Area : A
Height : H

$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$

$$\frac{q}{\varepsilon_0} = 2\,E\,A$$



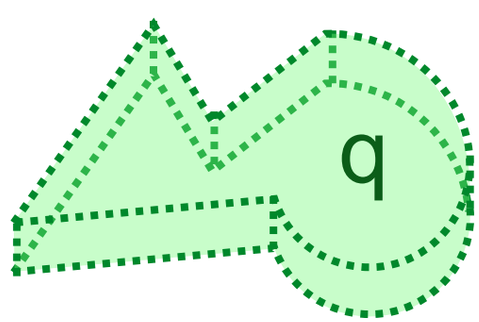
$$\eta = Q / A$$

$$\hookrightarrow q = \eta A$$

$$\frac{\eta \cancel{A}}{\varepsilon_0} = 2\,E\,\cancel{A}$$

$$E = \frac{\eta}{2\,\varepsilon_0}$$

If there was some thickness, T :
$q = \rho\, A\, T$



Infinite Surface of Charge

Limit if R → ∞

FROM LAST WEEK

$$\vec{E}_{plane} = \lim_{R \to \infty} \vec{E}_{disk}(R)$$

$$\vec{E}_{plane} = \lim_{R \to \infty} 2\pi K \eta \left(1 - \frac{z}{\sqrt{R^2 + z^2}}\right) \hat{k}$$

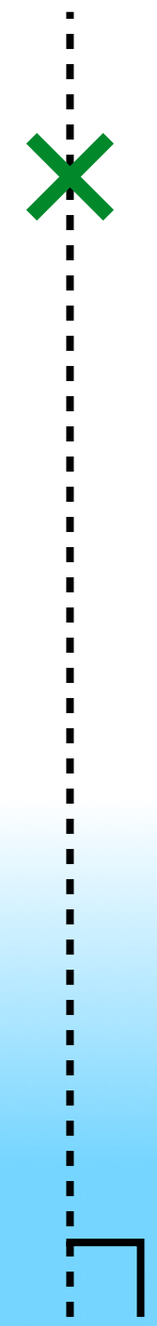
$$\vec{E}_{plane} = 2\pi K \eta \hat{k}$$

Compare to new equation

$$K = \frac{1}{4\pi\varepsilon_0}$$

$$\vec{E}_{plane} = \cancel{2}\cancel{\pi}\left(\frac{1}{\cancel{4}^{2}\cancel{\pi}\varepsilon_0}\right)\eta\hat{k}$$

$$\vec{E}_{plane} = \frac{\eta}{2\varepsilon_0}\hat{k}$$ *The same!*



Chapter 7

Electric Potential

The Electric Potential

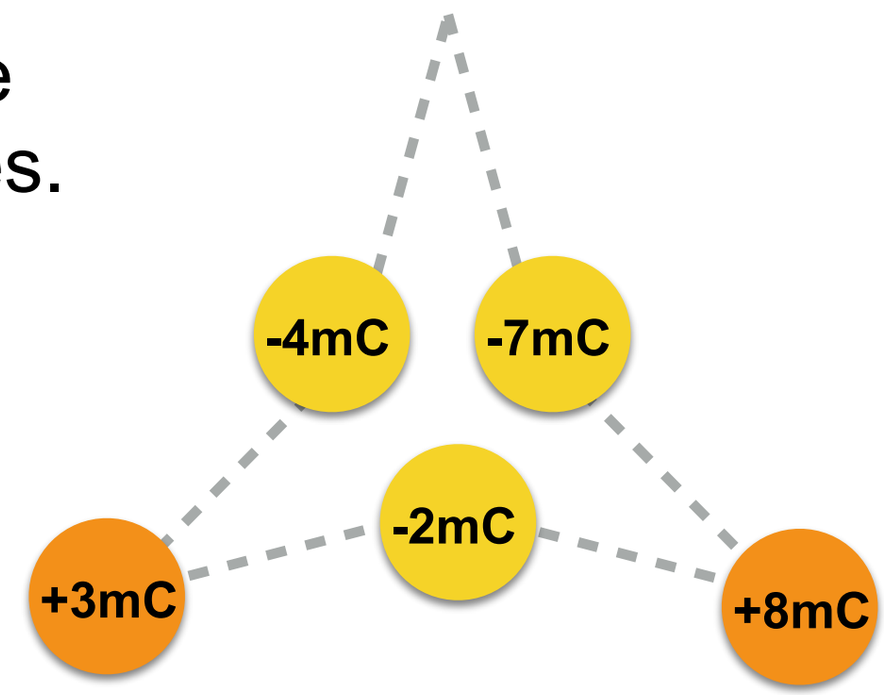
§ 7.2
Lecture 09

Adding Charge

Work is done to bring a +2 mC charge to the top of an arraignment of charges.

The system's potential energy changes by -6 mJ.

The electric potential at the top of the arrangement is…



1. 12 V
2. - 12 V
3. 3 V
4. - 3 V

$$V = \frac{U_q}{q} = \frac{-6mJ}{2mC}$$

Potential & Energy (& Force)

$$V = \frac{U_q}{q}$$

U_q is the energy of the *system of particles*

If only one particle moves, can treat it as the energy of that moving particle.

A bit of algebra… $U_q = q\,V$

Particle's Energy	=	Particle's Charge	×	Property of Environment

Compare to force : $\vec{F} = q\,\vec{E}$

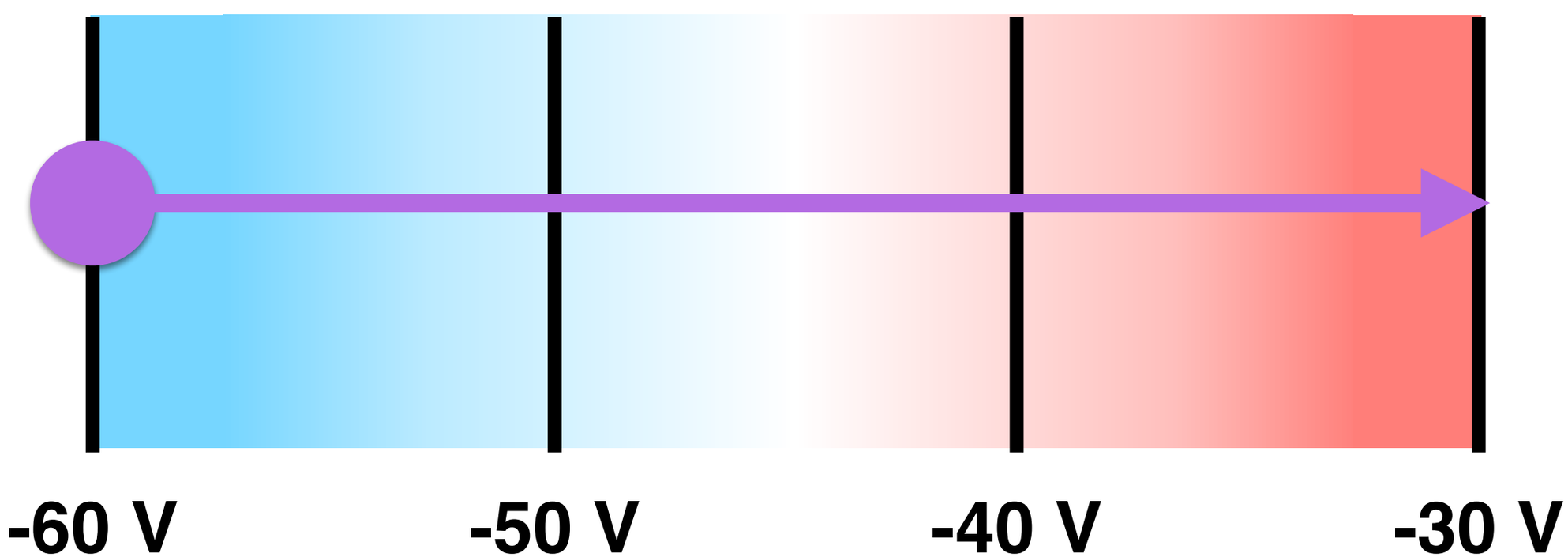
Force on Particle	=	Particle's Charge	×	Property of Environment

Changing Electric Potential

If a particle moves and the electric potential it experiences changes:

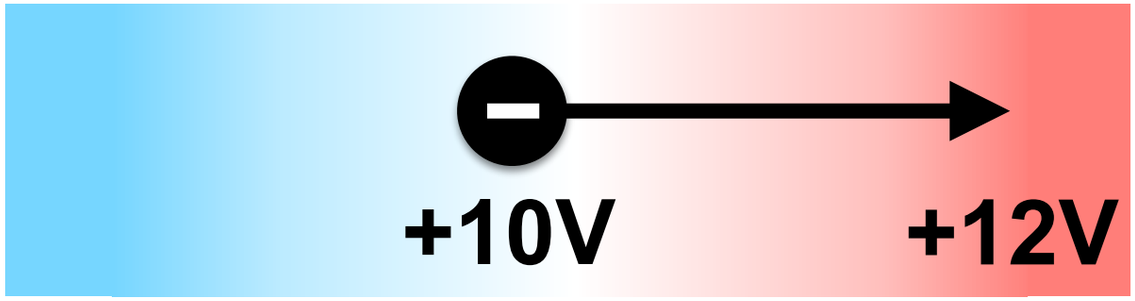
$$\Delta U_q = q\,\Delta V$$

Just like potential energy, where V=0 is arbitrary, only ΔV is important for motion.

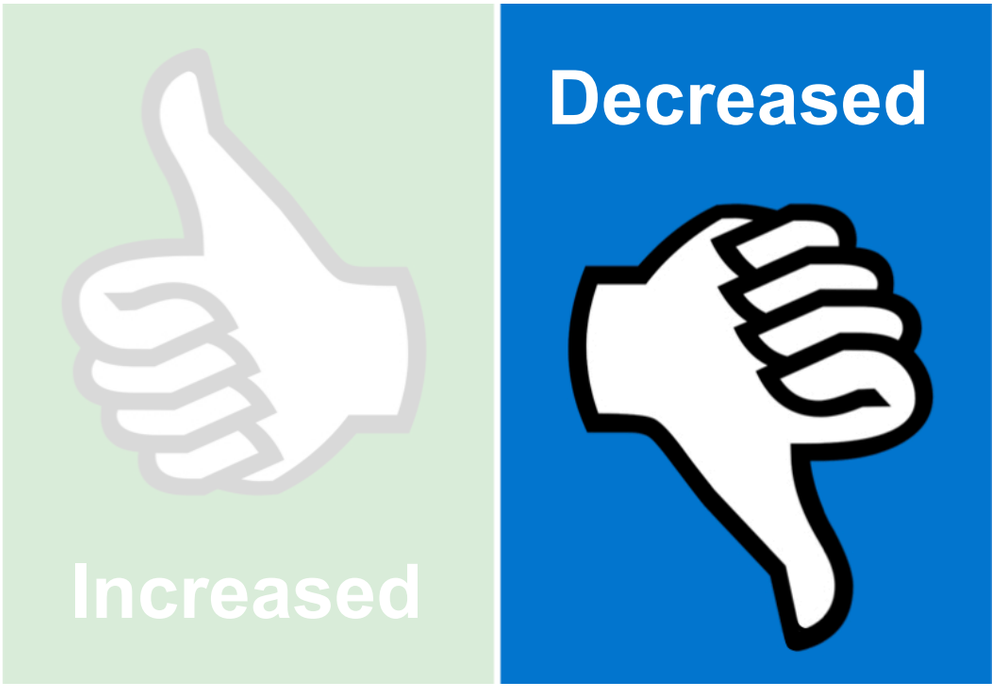


Moving Electron

An electron moves from +10V to +12V.



Has its *potential energy* increased or decreased?



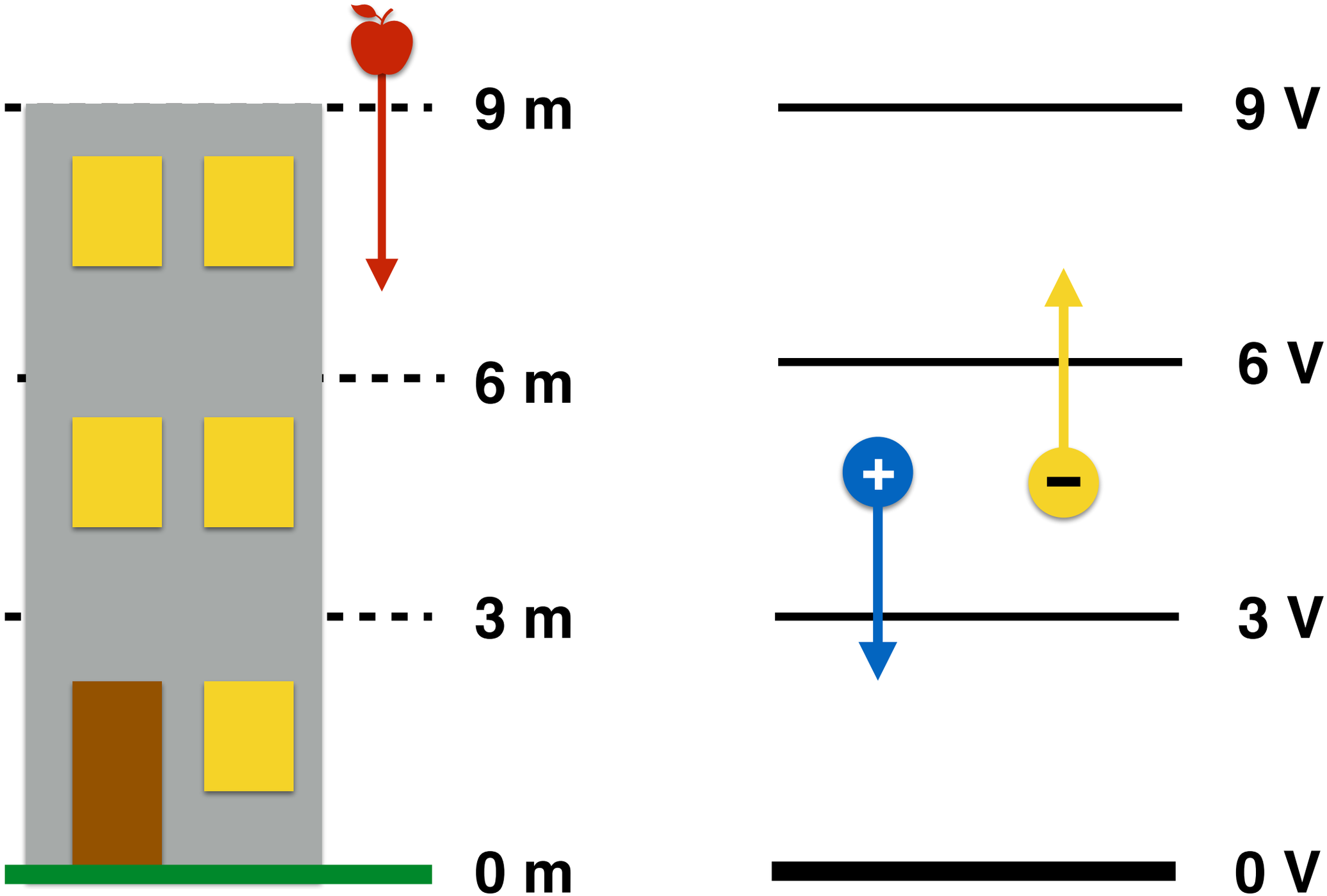
$$\Delta U_e = q\,\Delta V$$

Negative Positive

0. Don't wait on my response

It gains kinetic energy.

Electric Potenial & Charges



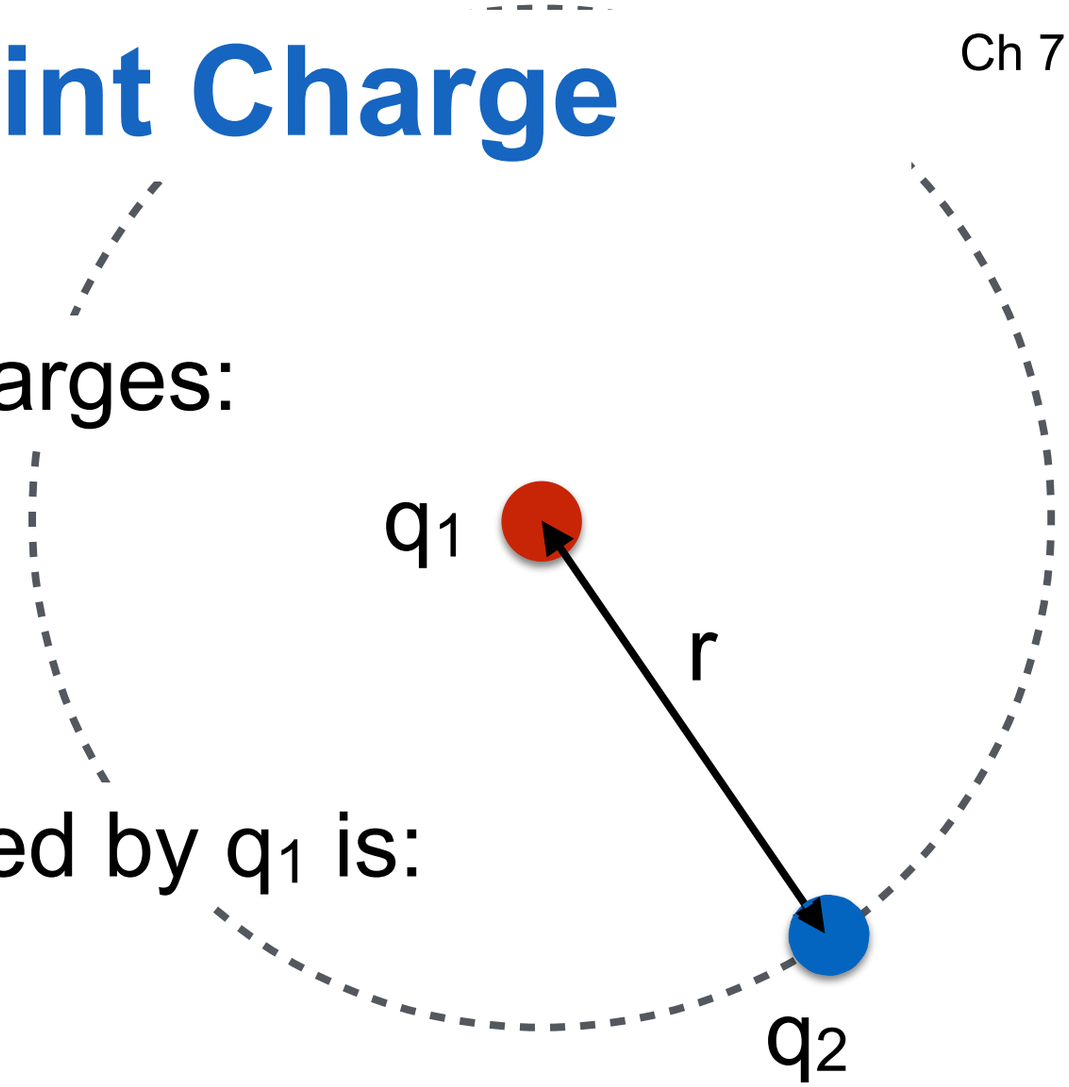
The Electric Potential of Charges

§ 7.3
Lecture 09

Potential of A Point Charge

We already saw for two point charges:

$$U_{elec} = \frac{K\, q_1\, q_2}{r}$$



At distance *r*, the potential created by q_1 is:

$$V(r) = \frac{U_{elec}}{q_2} = \frac{K\, q_1}{r}$$

Potential is radially symmetric around the charge

At very large distances from a charge: $\lim_{r \to \infty} V(r) = 0V$

Negative V $\rightarrow$ <u>Lower</u> potential than at V=0

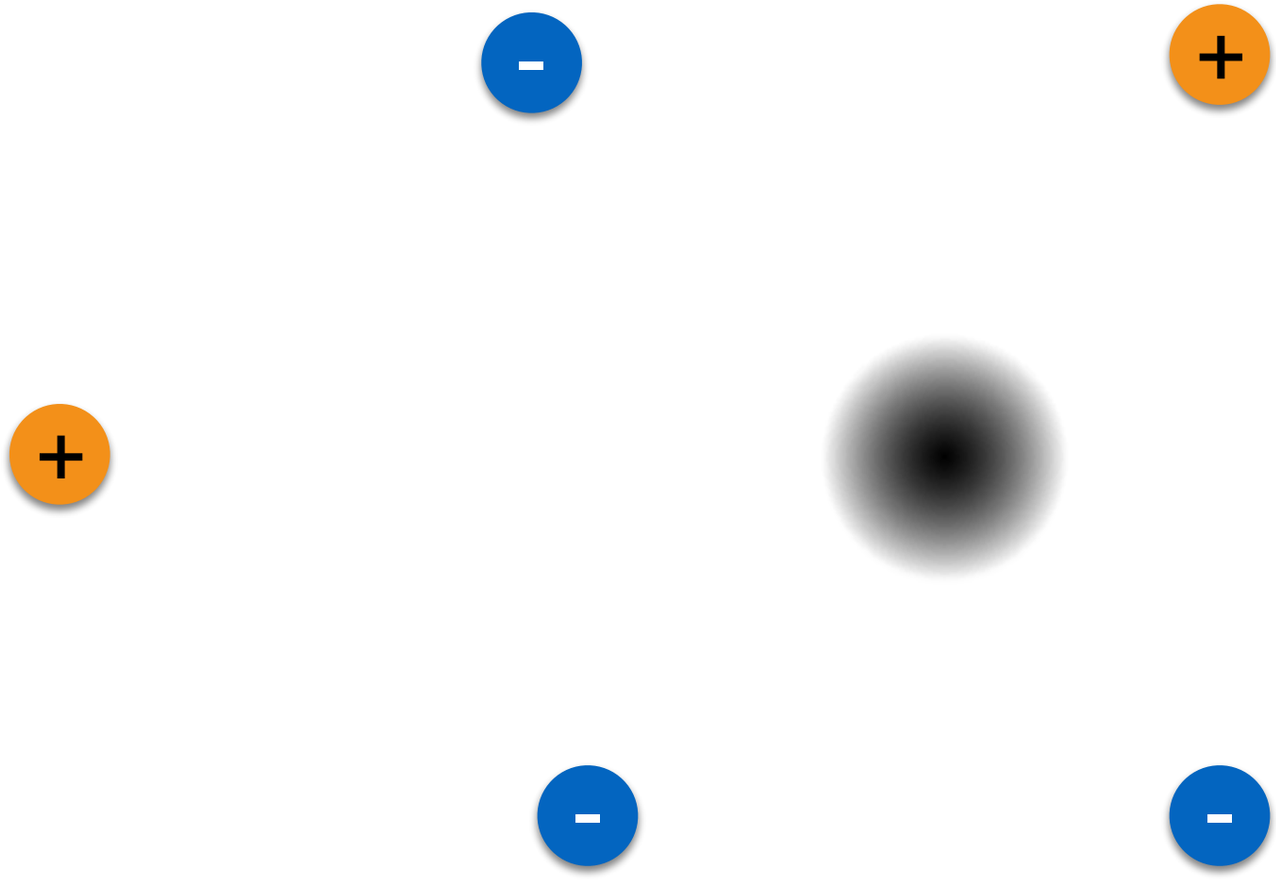
Potential of Many Point Charges

The potential at a point due to many changes is:

$$V = \sum_i V_i = K \sum_i \frac{q_i}{r_i}$$

Charge of each charge *i*

Distance from each charge *i*

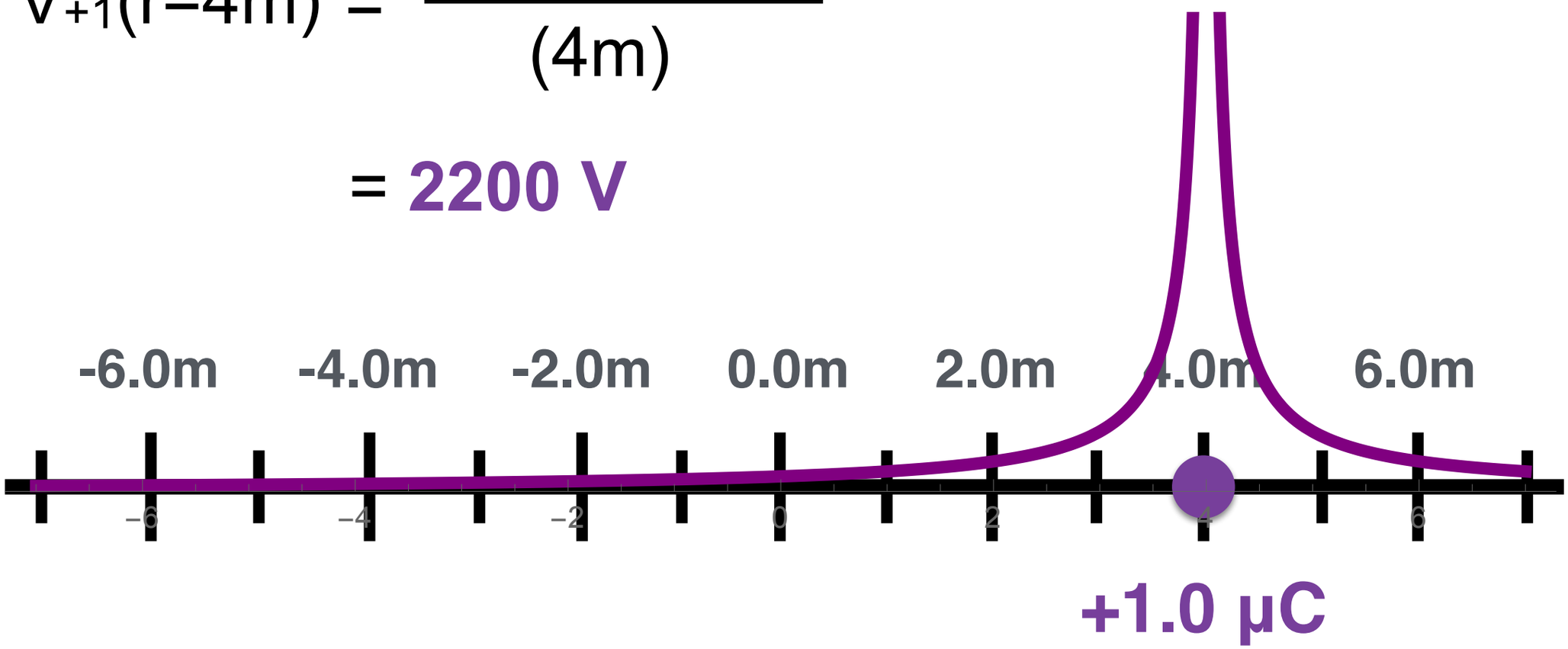


Potential from One Charge

What is the electric potential at x = 0m?

$$V_{+1}(r=4m) = \frac{K\,(+1\times10^{-6}C)}{(4m)}$$

$$= \mathbf{2200\ V}$$

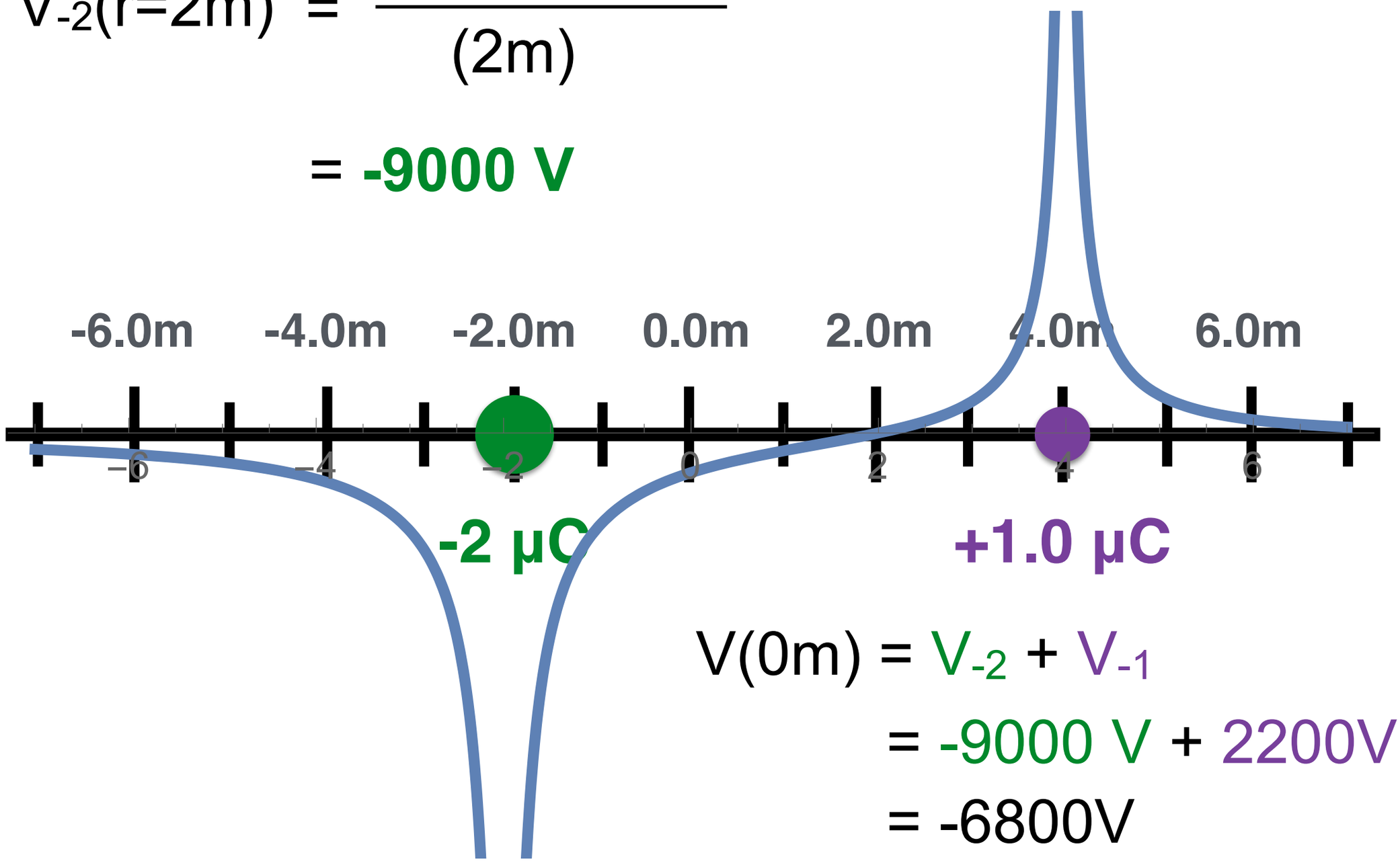


Potential from Many Charges

What is the electric potential at x = 0m?

$$V_{-2}(r=2m) = \frac{K\,(-2\times10^{-6}C)}{(2m)}$$

$$= \mathbf{-9000\ V}$$



$$V(0m) = V_{-2} + V_{-1}$$

$$= -9000\ V + 2200V$$

$$= -6800V$$

Pre-class Slides

Electric Potential Energy

§ 7.1
Lecture 09

Youtube Playlist

Electric Potential Energy

The Electrostatic force is *conservative*

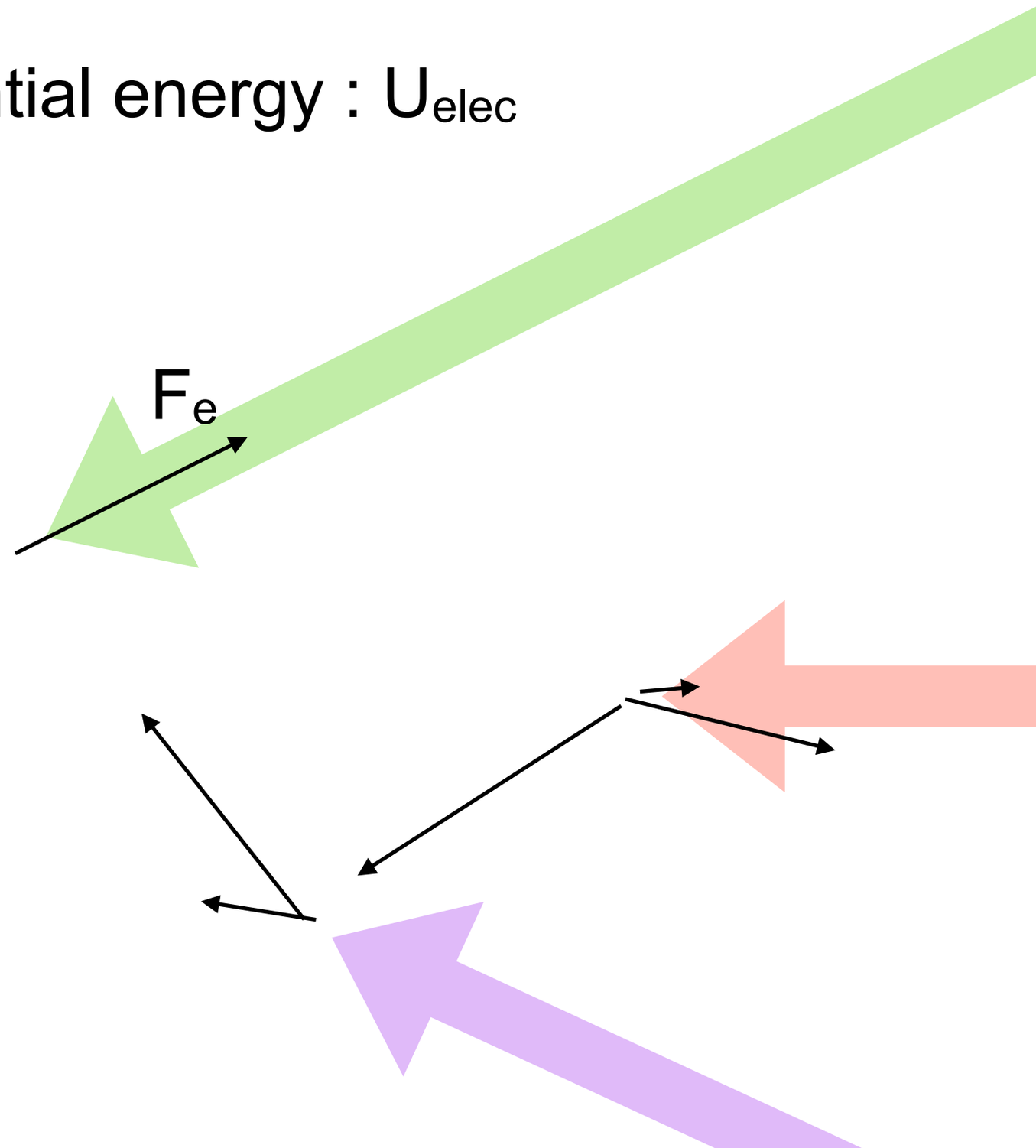
It has an associated potential energy : U_{elec}

$$\Delta U_{elec} = - \Delta KE$$

$$\Delta U_{elec} = - \sum W$$

Energy comes from the <u>work</u> done to position the charges

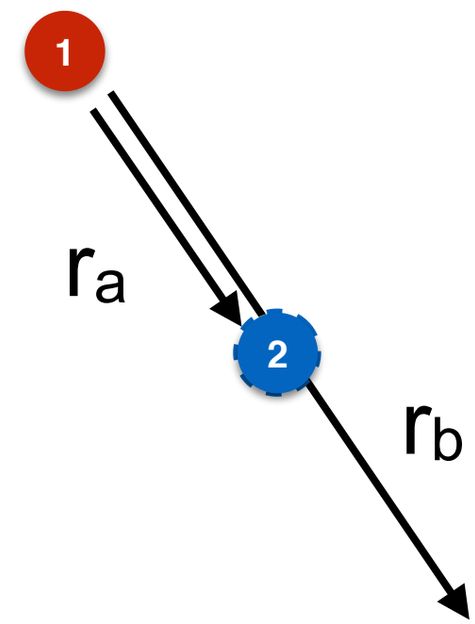
Addition of each charge changes U_{elec}



Electric Potential Energy

Point Charges

$$\Delta U_{elec} = -\int \vec{F}_e \cdot d\vec{r} = -\int_{r_a}^{r_b} \left(\frac{K\, q_1\, q_2}{r^2}\, \hat{r} \right) \cdot \left(dr\, \hat{r} \right)$$



$$U_{elec\ @\ b} - U_{elec\ @\ a} = \left(\frac{K\, q_1\, q_2}{r_b} \right) - \left(\frac{K\, q_1\, q_2}{r_a} \right)$$

$$U_{elec}(r) = \frac{K\, q_1\, q_2}{r}$$

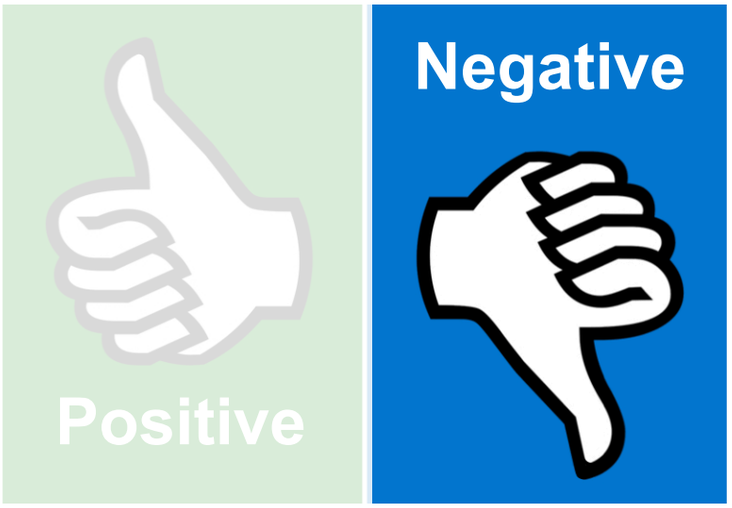
Potential Energy to arrange two charges a distance r apart

Electric Potential Energy

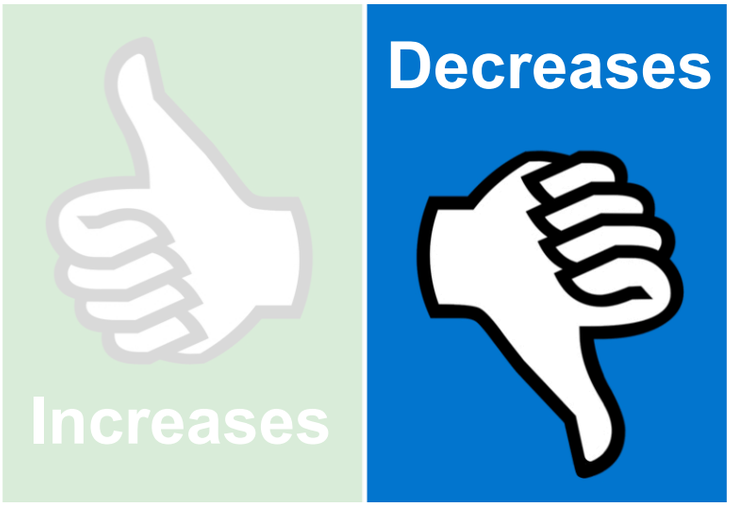
Two Charges

An electron and proton are separated by a small distance.

The potential energy between them is ….



If the two charges move closer together, their energy…



More negative

$$U_{elec}(r) = \frac{K\, q_1\, q_2}{r}$$

(+) (-)

r ↓

-3J → -15J

KE goes up!

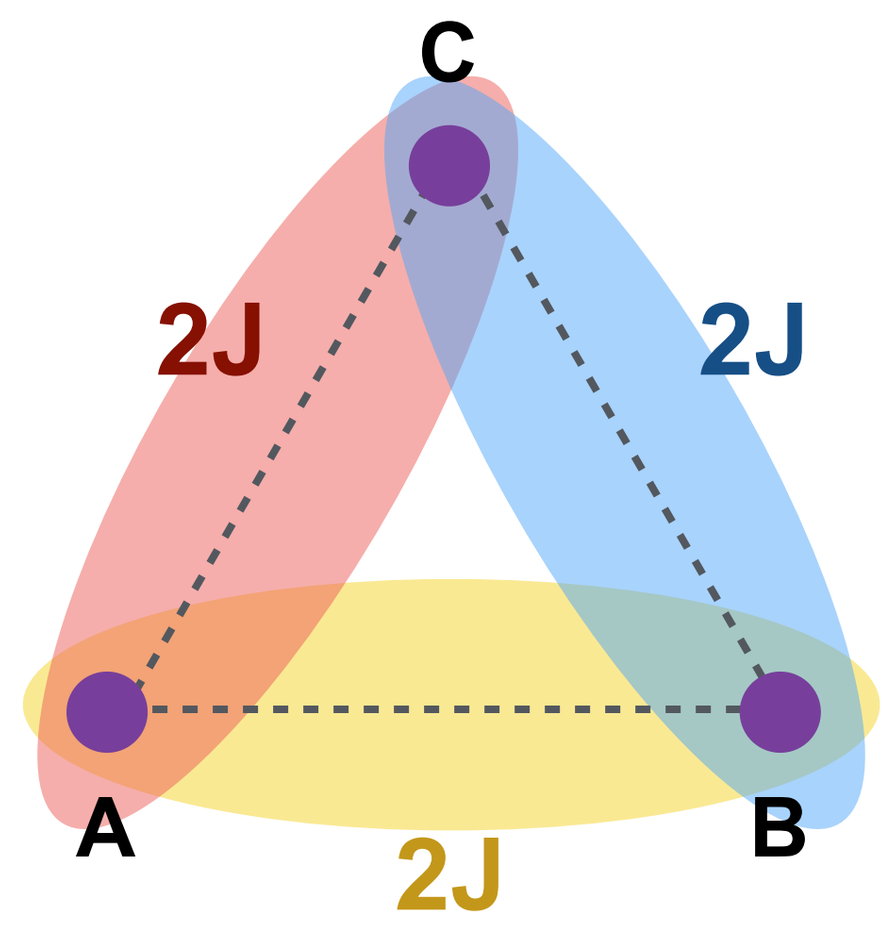
Equilateral Point Charges

Charges A, B and C all have the same charge and are separated by the same distance.

2J of energy is needed to bring together charges A and B.

What is the total energy of the system once charge C was added?

1. 2 J
2. 3 J
3. 4 J
4. 6 J
5. 12 J
0. Don't wait on my response



Non-equilateral Point Charges

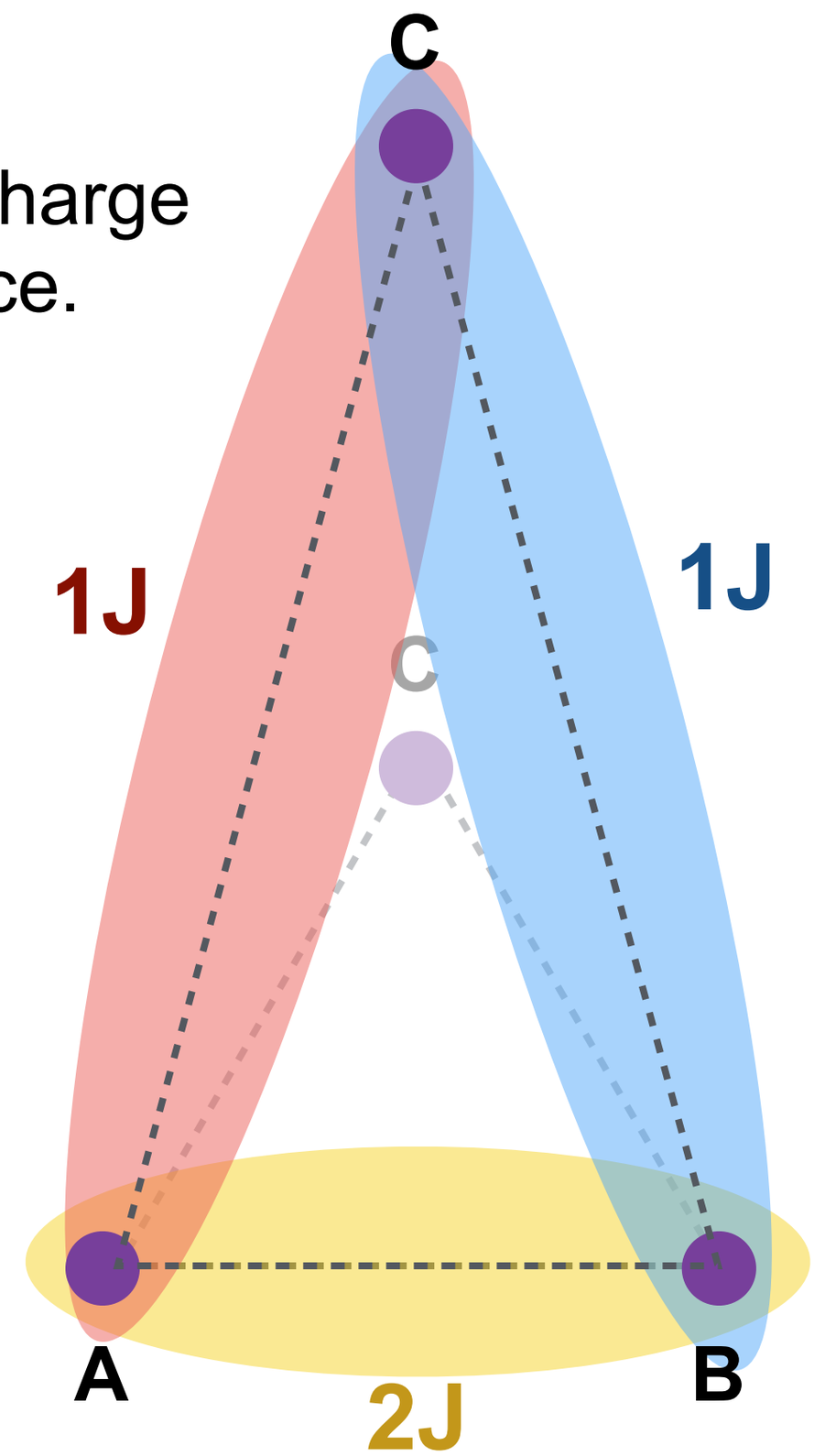
Charges A, B and C all have the same charge and were separated by the same distance.

The total system had an energy of 6J.

What would the energy be if C were twice as far away from A and B?

1. 2 J
2. 3 J
3. 4 J
4. 6 J
5. 12 J
0. Don't wait on my response

$$U_{elec} \propto \frac{1}{r}$$

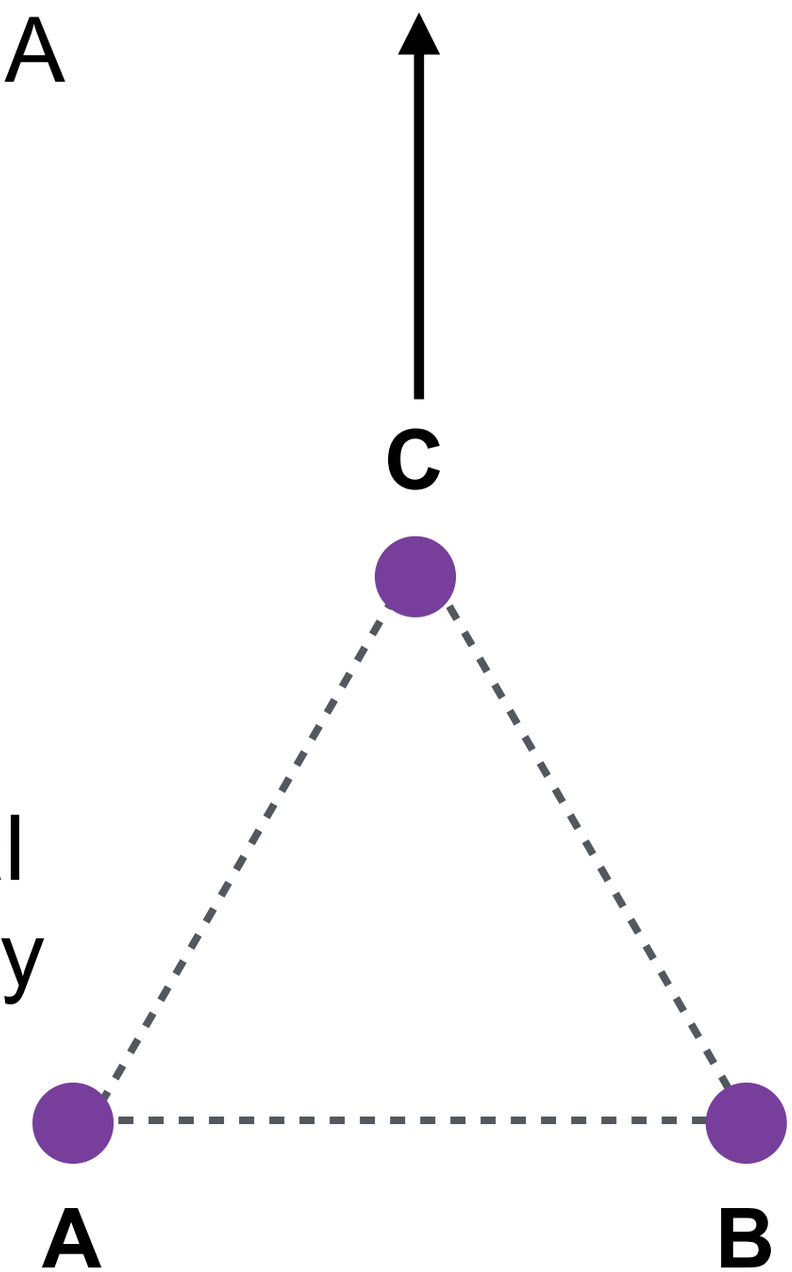


Moving Point Charges

As charge C moves away from charges A and B, the system <u>loses energy</u>.

That energy can become kinetic energy for charge C!

Or, if charge C moves closer, its kinetic energy can be transformed into potential energy (just like objects moving vertically in the air).



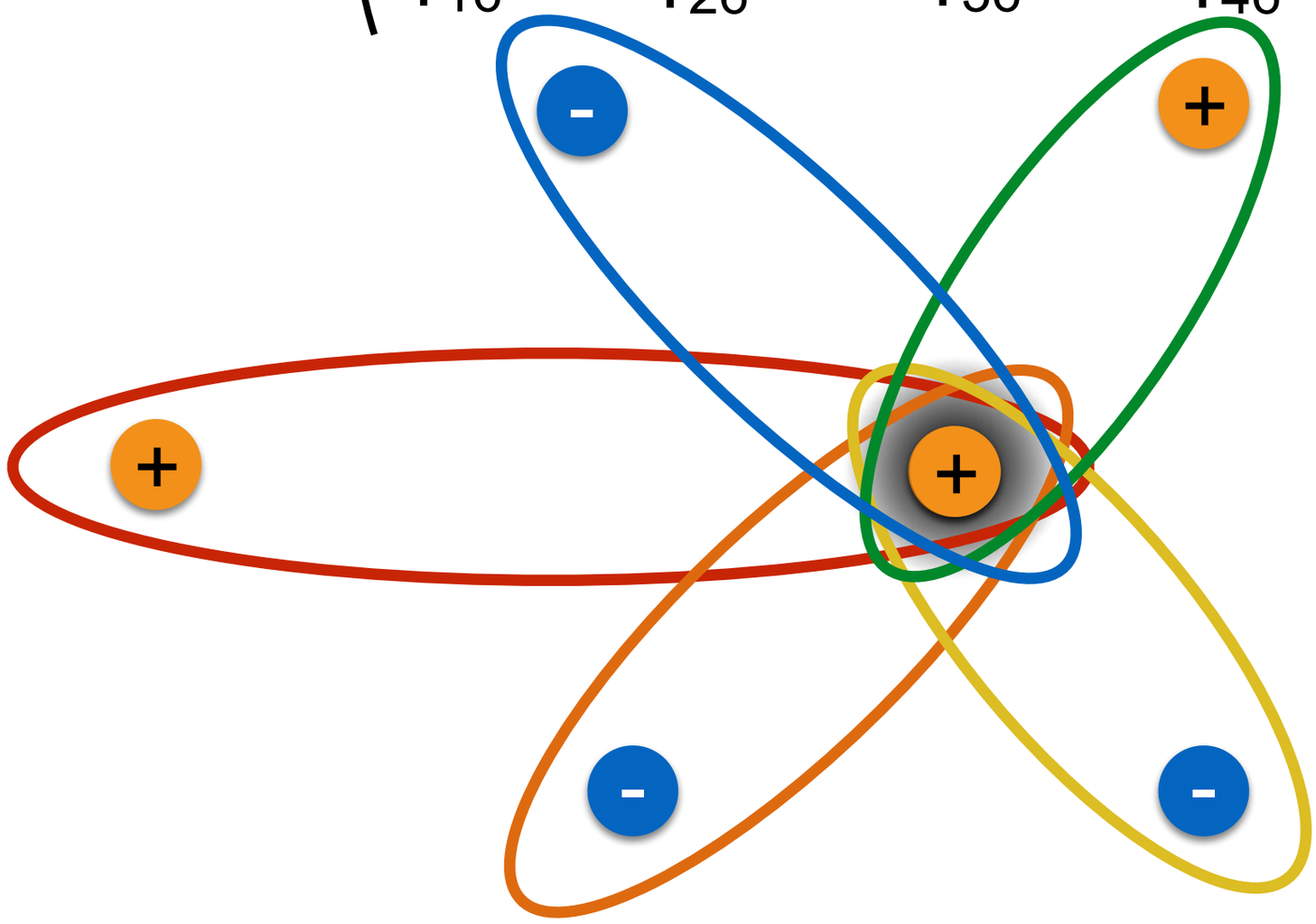
The Electric Potential

§ 7.2
Lecture 09

Energy for An Extra Charge

$$U_{5\text{ charges}} + \frac{Kq_1q_6}{r_{16}} + \frac{Kq_2q_6}{r_{26}} + \frac{Kq_3q_6}{r_{36}} + \frac{Kq_4q_6}{r_{46}} + \frac{Kq_5q_6}{r_{56}}$$

$$= U_{5\text{ charges}} + \left(\frac{Kq_1}{r_{16}} + \frac{Kq_2}{r_{26}} + \frac{Kq_3}{r_{36}} + \frac{Kq_4}{r_{46}} + \frac{Kq_5}{r_{56}}\right)\text{-7nC}$$



The Electric Potential

Electric potential tells electric potential energy per charge <u>at a location</u>.

Units : J/C or Volts

Energy gained by the system due to a charge at position (x,y,z)

$$V = \frac{U_q}{q}$$

Electric Potential at position (x,y,z) created by source charges

Charge at the position